

Money Tree trade audit

Every trade from the last four sessions (22–25 Sep 2026) replayed against the strategy code and `spec.strategies.md` . Pattern checks go back to 14 Sep for `breakout_5m`. Bars, quotes, orders and fills come from Alpaca: IEX 5/10-minute bars and IEX quotes (the bot's intraday feed), and SIP daily bars (the bot's daily feed).

Orders read **855** Fills read **1,381** Code 22–24 Sep **a1d0b67** Code 25 Sep **6b22302**

Strategies traded **breakout_5m** · **breakout_10m** · **daily_sma** · **daily_tfb**

13
Confirmations

6
Errors

5
Issues

6
Queries

8
Improvements

Answers to the nine questions

QUESTION	VERDICT	DETAIL
Was each entry valid?	MOSTLY	All 40 entries (15 breakout, 25 daily) passed the coded setup at signal time. SAIL (breakout_10m) broke the extension limit. VIAV, PL and MSFT filled outside the 1-5% stop band because entries ran late (E6, I1, I2).
Was the correct timeframe used?	YES	breakout_5m read 5Min bars, breakout_10m read 10Min bars, daily read 1Day SIP bars. All entries fell inside each entry window (C2).
Was the correct OR/high/low calculated?	YES	For all 15 breakout entries on 25 Sep, the resting stop matches the replayed opening-range stop to the cent (C1).
Was volume confirmation correct?	YES	All 15 met the multiple. The lowest were PSKY at 1.37× and GME at 1.38×, against 1.3× required (C3).
Was entry extension respected?	NO, ONCE	SAIL entered at 21.53. The limit was 21.70, so the entry is 0.64× the range beyond the low against a 0.25× maximum (E6). MSFT passed. breakout_5m has no limit (none).
Did the exit happen for the correct coded reason?	NO	6 of 13 breakout_5m holdings on 25 Sep took a "target" exit at a price the market never traded (E1, E2). 13 of 17 daily exits on 25 Sep had no coded exit signal (E5). All daily exits on 23–24 Sep were correct (C7).
Did pending orders count toward caps?	YES	On every breakout day, 10 entries went out within 5 seconds and the 11th was held back. daily_sma peaked at 10 and the portfolio at 20 (C8).
Did any strategy enter a stock another strategy already held?	NO	Zero overlaps across the whole account history. MSFT changed hands on 25 Sep: daily_sma sold at 09:30:37 and breakout_10m bought at 10:17:57 (C9).
Did actual execution match the strategy register?	PARTLY	Codes, pause flags and the selection all match (C12). Three untracked positions (MARA, SAM, SUNB) sit outside every strategy, and on 22–24 Sep they filled the 20-position book, which blocked breakout_5m (E3, E4).

ERRORS

Behaviour that breaks a coded rule

E1 Profit targets fire on IEX quote midpoints the market never traded

The bot's price comes from Lumibot's `get_last_price`, which returns the midpoint of the latest IEX quote (`lumibot/data_sources/alpaca_data.py:1303`). IEX quotes on these names are often several percent wide, so the midpoint jumps past the first target.

`Breakout.manage` then sells the first share of the position.

25 SEP	SIDE	ENTRY	TARGET 1	BEST REAL PRICE BEFORE EXIT	IEX BID / ASK AT EXIT	IEX MID	EXIT FILL
JCI 09:57	long	148.756	151.21	149.265	149.06 / 156.45	152.755	149.162
EQPT 09:57	long	17.03	17.555	17.16	17.10 / 19.42	18.26	17.10
S 10:00	short	23.26	22.795	23.21	22.18 / 23.30	22.74	23.27
CXW 10:13	short	32.767	31.997	32.70	28.56 / 32.84	30.70	32.77
IONS 10:54	long	44.68	45.475	44.79	44.64 / 46.71	45.675	44.65
QCOM 11:08	long	200.188	206.59	203.72	203.43 / 210.00	206.715	203.458

- From 14 to 25 Sep, **36 of 46** breakout partial exits fired before the SIP high or low reached target 1. Only 10 were real.
- The two real ones on 25 Sep were `breakout_10m`: MSFT reached 516.91 against a 516.19 target, and SAIL reached 20.75 against 20.75.

E2 Each phantom target is followed by an instant breakeven stop-out

After target 1, the stop moves to the entry price (`breakout.py:211`). The real price is still at or below entry, so `protect` reads the stop as already passed and sells the rest at market (`portfolio.py:505-514`). A trade that should run until a real target or the close ends within minutes, near breakeven.

- The rest sold 3 seconds to 6 minutes later: JCI 09:59:02, EQPT 09:59:00, S 10:00:05, CXW 10:19:00, IONS 10:56:00, QCOM 11:09:02.
- SAIL (`breakout_10m`) at 15:53:02: the IEX mid was 21.45 (bid 20.77 / ask 22.13) against a 20.88 trail stop. The bot bought to cover at market at 20.78, below the stop. The resting stop at the broker would not have triggered.

E3 Stops placed after a position is closed open new, unmanaged positions

When an exit fills, `_fill` and `_resync_stops` read Lumibot's cached position quantity. That cache still shows the old size, so the bot places a fresh stop on a flat position. The broker files it as `buy_to_open` / `sell_to_open`. A duplicate exit does the same thing.

EVENT	ORDER	RESULT
09 Sep BKNG	buy stop 47 @ 173.50 after the short was covered	Filled 47. It opened a long that was closed by hand on 14 Sep.
18 Sep SUNB	buy stop 32 @ 73.14 after cover	Filled. 32 long is still open.
21 Sep SAM	second market buy 13 one minute after the first cover	Filled. 13 long is still open.
21 Sep MARA	buy stop 177 @ 13.27 after cover at 15:54	Filled 15:57. 177 long is still open.
25 Sep SAIL, GME, PURR	buy stops 2, 1 and 7 placed 15:54-15:56 after each cover	Expired unfilled at 16:00.

- 17 stop orders since 9 Sep had an opening intent. 3 of them filled (BKNG, SUNB, MARA). SAM came from a duplicate market exit.
- Open now, with no strategy, no stop and no exit rule: **MARA 177 long** (−\$130.98), **SAM 13 long** (−\$80.86), **SUNB 32 long** (+\$60.16).

E4 The stray positions shut breakout_5m out for three sessions

`enter` counts every broker position against `RISK__POSITIONS_MAX = 20` (`portfolio.py:447`). From 22 to 24 Sep the book held 17 daily positions plus MARA, SAM and SUNB, which is exactly 20. `breakout_5m` was selected and unpaused, but it placed zero orders on those three days. It had placed 10 entries by 09:54 on each of the six sessions before.

E5 13 of 17 daily exits on 25 Sep have no coded reason

All 17 daily holdings were sold at 05:39:41-42 ET, before the open, with `mt-s` and `mt-t` codes. The only code path for those codes is `Daily.manage`: stop, signal or earnings. The earnings branch cannot run before the open, and `daily_tfb` does not heed earnings. On the 24 Sep close:

VALID (SIGNAL)	NO SIGNAL, NO STOP BREACH
AKAM (RSI 49.25)	<code>daily_sma</code> : ESTC, MRK, OGN, OMER, PFE, VEEV
HELP (<code>close 12.59 < SMA20 12.81</code>)	<code>daily_tfb</code> : COIN, CRWD, META, MRNA, MSTR, PLTR, WBD
MSFT (<code>497.93 < 499.49</code>)	
U (<code>41.42 < 42.47</code> , RSI 47.2)	

The exits coincide with the 25 Sep deploy (PR #46 at 05:26 ET, #47 at 05:40 ET). The 13 exits netted +\$137 combined. See Q1.

E6 SAIL (breakout_10m) entered past its extension limit

The opening range was 21.81-22.25, so the limit is $21.81 - 0.25 \times 0.44 = \mathbf{21.70}$. The signal close was 21.54, the IEX mid at submission was 21.56, and the short filled at 21.53. `is_overextended` at the running revision returns true for any price below 21.70, so the entry should have been skipped. See Q2.

ISSUES**Allowed by the code, off from the written intent****I1 Entries go out 4-28 minutes after the scan that found the signal**

The first breakout entries land at 09:53-09:54 on every session. The scan that saw the break ran at 09:40 or 09:45. On 25 Sep, entries lagged their scan by 13.8 minutes (COST, QCOM, JCI, CSGP, VIAV, PSKY, EQPT, S), 8.9 minutes (GME, PURR), 9.1 minutes (IONS) and 7.1 minutes (CXW). MSFT (10m) went out at 10:17:57 after a 09:50 break. Because of the delay, fills sit far past the breakout level: CSGP 3.8×, GME 3.7×, VIAV 3.4× and QCOM 2.2× the range width. The spec says "enter at the next open".

I2 Stop bounds are checked on the signal close, not the fill

`is_setup_ready` tests the 1-5% stop distance against the signal-bar close only. At the fill price: VIAV risked **6.15%** (above the 5% maximum), PL **0.78%** and MSFT **0.48%** (below the 1% minimum). MSFT filled at 511.30, back inside its 497.35-512.69 range.

I3 Sizing, extension and the "quote through stop" guard use the same midpoint

Long quantities equal $\$100 \div \text{the bot's price}$, so that price can be read back from each order. It matches the IEX mid, not the trade price: EQPT 16.69 against a 17.03 fill, IONS 44.17 against 44.68, COST 907.95 against 920.11. At the fill price, COST (\$101.34), EQPT (\$102.04), IONS (\$101.15) and VIAV (\$100.71) exceed `RISK__NOTIONAL_USD_MAX = 100`.

I4 Heavy stop-order churn

376 stop orders since 14 Sep; 328 were canceled. During partial fills each fill replaces the stop. EQPT got four stops in 11 seconds for 1, 0.99, 4 and 5.99 shares. SAIL got four in 6 seconds.

I5 Volume confirmation reads IEX volume only

`BARS__INTRADAY_FEED = iex`. IEX carries a small share of consolidated volume, so the ratio compares one venue's volume to its own history. The ratio stays internally consistent, but thin names swing more. PSKY (1.37×) and GME (1.38×) cleared the 1.3× bar by less than 0.1.

CONFIRMATIONS**Behaviour that matches the code and spec**

c1 Opening range, levels and stop: exact on all 15 breakout entries (25 Sep)

SYMBOL	STRAT	DIR	OR HIGH	OR LOW	SIGNAL BAR	SIGNAL CLOSE	RANGE %	STOP % (SIGNAL)	VOL ×	MODEL STOP	BROKER STOP
COST	5m	long	909.05	884.765	09:35	913.00	2.66	1.10	7.71	902.97	902.97
QCOM	5m	long	196.36	194.62	09:35	199.11	0.87	1.60	1.66	195.92	195.92
JCI	5m	long	147.395	146.31	09:35	149.40	0.73	1.52	4.08	147.12	147.12
GME	5m	short	24.89	24.685	09:40	24.48	0.84	1.05	1.38	24.74	24.74
PURR	5m	short	13.96	13.31	09:40	13.09	4.97	2.92	2.18	13.48	13.48
CSGP	5m	long	27.755	27.60	09:35	28.00	0.55	1.01	1.46	27.71	27.71
VIAV	5m	long	38.20	37.52	09:35	38.86	1.75	2.14	1.77	38.03	38.03
PSKY	5m	short	10.255	10.105	09:35	10.025	1.50	1.17	1.37	10.15	10.15
EQPT	5m	long	16.80	16.34	09:35	16.90	2.72	1.27	18.55	16.68	16.68
S	5m	short	23.89	23.455	09:35	23.205	1.87	1.55	2.79	23.57	23.57
IONS	5m	long	44.245	43.865	09:50	44.71	0.85	1.25	2.31	44.15	44.15
PL	5m	long	17.94	17.57	09:55	18.16	2.04	1.72	3.36	17.84	17.84
CXW	5m	short	33.845	33.08	10:00	32.93	2.32	1.04	2.00	33.28	33.28
MSFT	10m	long	512.69	497.35	09:40	515.42	2.98	1.27	3.64	508.85	508.85
SAIL	10m	short	22.25	21.81	10:00	21.54	2.04	1.76	2.30	21.92	21.92

Long stop = low + 0.75 × width; short stop = low + 0.25 × width. All ranges were ≥ 0.4%, all stop distances fell in 1–5% at the signal close, and every signal was 1 bar old at the first scan that saw it (maximum 2).

c2 Timeframes and windows

breakout_5m ranges come from the 09:30 5Min bar, and entries went out between 09:53 and 10:12 (scan window 09:35–10:30). breakout_10m ranges come from the 09:30 10Min bar, and entries went out at 10:17 and 10:26 (window 09:40–10:30). Daily signals use completed 1Day SIP bars through the prior session. Daily entries were submitted at 09:30:09–09:34:00, which is "the next open, else the next permitted iteration".

c3 Volume confirmation

For all 15 breakout entries, cumulative volume to the signal bar ÷ the 20-session mean at the same clock time was ≥ the variation multiple (1.3× for 5m, 1.5× for 10m). The range ran from 1.37× (PSKY) to 18.55× (EQPT).

c4 All 25 daily entries (22-24 Sep) pass the coded setup

daily_sma (18 entries: MSFT twice, MRK, AKAM, U, ZETA, ESTC, NSIT, ARX, LFST, HELP, AMGN, FNV, ELF, OMER, PFE, VEEV, OGN): each has close > SMA50 > SMA200, RSI $\geq$ 50, ADX $\geq$ 25, a cross above SMA20, and a close above the prior close. daily_tfb (7 entries: META, WBD, MSTR, PLTR, MRNA, CRWD, COIN): each has 20-session turnover > \$20M, close > a rising SMA50 (3-session lag), ADX $\geq$ 20, and a close above the prior high. SPY was above its SMA20 on every entry day. Every symbol cleared the \$5 price and \$20M turnover universe filters. Sizes match 0.1% of equity $\div$ (1.5 or 2.0 $\times$ ATR), capped at 5% of equity. ARX and OGN hit the 5% cap.

c5 Daily exits on 23-24 Sep are exact

23 Sep: NSIT, MSFT, ZETA and LFST closed below SMA20. 24 Sep: ARX, AMGN, FNV and ELF closed below SMA20. The replay finds these 8 signals and no others among the 17 holdings on each day, so no exit was missed. All 8 were submitted at 09:30:07-09:30:14 and filled at the open.

c6 Correct breakout exits on 25 Sep

- Close lead (15:54, 6 minutes before the close): COST, VIAV, GME, PURR, CSGP.
- Resting stop filled at the broker: PL at 17.82 (stop 17.84) and PSKY at 10.152 (stop 10.15).
- MSFT (10m): real target 1 sold half at 516.14. The stop then went to breakeven at 511.29 and trailed 512.92 $\rightarrow$ 513.18, never widening, and filled at 513.13.
- SAIL (10m): real target 1 covered 2 at 20.72. The stop went to breakeven at 21.53, then trailed to 20.88.

c7 Stops never widened

Across every same-day sequence of closing stop orders since 14 Sep, long stops only rose and short stops only fell. Zero violations.

c8 Pending orders count toward caps

`holding_count` adds pending entries (`portfolio.py:193-200`), and `enter` adds pending to the position count and to gross exposure. Observed peaks, pending included: the breakout family hit exactly 10 on all seven breakout sessions from 14 to 25 Sep (family cap 10 at the running revision), with 10 orders inside 5 seconds each time. `daily_sma` peaked at 10 (cap 10 at the running revision), `daily_tfb` at 7, the whole book at 20.

c9 No cross-strategy overlap

Across all 855 orders, no strategy submitted an entry while another strategy held or had a pending order in the same symbol.

C10 Direction, sizing and whole-share shorts

All 6 shorts on 25 Sep used whole shares (GME 4, PURR 7, PSKY 9, S 4, CXW 3, SAIL 4). Partial short exits rounded down (CXW 3 → 1, SAIL 4 → 2). Longs were fractional and sized to the \$100 notional cap.

C11 One entry per asset per session

No symbol had two entries from the same strategy on the same day. MSFT's two daily_sma entries (22 and 24 Sep) each followed a fresh SMA20 cross.

C12 Register and pause flags match execution

22–24 Sep (a1d0b67): daily_sma and daily_tfb were unpaused and traded; breakout_10m was paused and placed zero orders. 25 Sep (7aa3ef2 config): daily_sma and daily_tfb were paused and placed only exits; breakout_10m was unpaused and placed 2 entries. breakout_15m (f) and daily_20sma (w) were not yet in the deployed registry, and no f or w orders exist. Every mt- order code maps to its strategy.

C13 Realized P&L for the audited sessions

breakout_5m on 25 Sep: +\$2.34 over 13 trades. breakout_10m on 25 Sep: +\$3.77 over 2 trades. daily_sma 22–25 Sep: −\$373.16. daily_tfb 22–25 Sep: +\$174.61. Open stray positions: −\$151.68 unrealized.

QUERIES

Needs data this session could not read

Q1 What sold all 17 daily holdings at 05:39 ET on 25 Sep?

The bot log for 09:39 UTC on money-tree-bot will show whether Daily.manage ran and which frames it read. The code keeps holdings only in memory (spec.md defers restarts), so a freshly started process has no holdings to sell.

Q2 What value did the bot run for BREAKOUT_10M__ENTRY_EXTENSION_MAX on 25 Sep?

SAIL passes only if the live value was none or ≥ 0.64 . Railway variables were not read in this session; access was declined.

Q3 Which revision ran each session?

This audit assumes a1d0b67 for 22–24 Sep and 6b22302 for 25 Sep, based on merge times. The strategy logic is identical between them except for pause flags and the \$100 notional cap. The Railway deployment history confirms it.

Q4 Earnings blocks and the common-stock filter

daily_sma heeds earnings within 5 days, and the universe intersects Finnhub common stocks. No Finnhub key was available, so both checks are unverified for the 18 daily_sma entries.

Q5 Were the right 10 chosen when more than 10 qualified?

Candidates are ranked by last-session turnover before the cap cuts them off. Verifying the choice needs a full-universe replay (all tradable stocks, 390 days). This audit confirms that each chosen symbol qualified, not that it outranked the ones left out.

Q6 MSFT (breakout_10m) timing

The break was visible at the 09:50 scan. At the 10:10 scan the signal is 3 bars old, above the 2-bar maximum. The entry at 10:17:57 fits only if the 10:00 iteration ran for 18 minutes, or if the live IEX bars differed from the stored ones.

IMPROVEMENTS

#	CHANGE	FIXES
P1	Read the latest trade price (or SIP NBBO) instead of the IEX quote midpoint, and reject quotes whose spread exceeds a set fraction of price.	E1, E2, I3
P2	Rest target orders at the broker (limit or bracket/OCO) so targets trigger only on real trades.	E1, E2, I4
P3	Read broker positions, not Lumibot's cached quantity, before placing any stop, and cancel every stop for a symbol once it is flat.	E3
P4	Flag and alert on broker positions that no holding owns, then close or adopt them. MARA, SAM and SUNB are open now.	E3, E4
P5	Re-run the stop-band and extension checks at the entry price just before submitting.	I1, I2
P6	Prefetch the universe and the 45-day confirmation bars before 09:35, so entries go out on the bar they are meant for.	I1, Q6
P7	Log each exit's reason and inputs (price, target, stop) to the state events. Order reasons are tagged in the code since #54 but were not live for these trades.	E5, Q1
P8	Deploy only while flat, or rebuild holdings from broker positions and order codes on start.	E5

Plain-language summary

The bot picks its trades correctly. It draws the right price ranges, applies the right volume tests, sets its stop-losses in the right place, stays under its position limits, and never lets two strategies buy the same stock.

The problems are in how it sells. It reads a "price" that is the midpoint of a thin, lopsided quote. That fake price often looks like a profit target was hit when it was not, so the bot sells early, near break-even. It also sometimes leaves a stop order behind after a trade is closed, and three of those have turned into leftover stock positions (MARA, SAM, SUNB) that nothing manages. Those leftovers filled the account's 20-position limit and blocked the breakout strategy for three days.

On 25 Sep before the open, the bot also sold every daily position, and most of those sales had no rule behind them. The bot's logs are needed to find out why.